Practical 3A

Demonstrate CRUD operations with database using RESTful Web Service

What you will build

A simple **Player Database API** where you can:

- Create Player
- Read Player
- Update Player
- Delete Player

1️⃣ Software Required

Install these first:

1. **NetBeans IDE (8.x recommended for this journal)**
2. **JDK 8**
3. **GlassFish Server** (comes with NetBeans)
4. **Java DB (Derby Database)** (already inside NetBeans)

2️⃣ Start Database

1. Open **NetBeans**
2. Go to **Services tab**
3. Expand **Databases**
4. Right click **Java DB**
5. Click **Start Server**

If server starts → correct ✅

3️⃣ Create Database

1. Right click **Java DB**
2. Click **Create Database**

Fill this:

Database Name : playerDB

User Name : root

Password : root

Click **OK**

Database created ✅

4️⃣ Create Table

1. Expand

Databases

→ jdbc:derby://localhost:1527/playerDB

2. Right click **Tables**
3. Click **Execute Command**

Now paste this SQL:

```
CREATE TABLE player(
id INT PRIMARY KEY,
name VARCHAR(50),
country VARCHAR(50)
);
```

Click **Run**

Table created ✅

5️⃣ Create REST Project

1. Click **File**
2. Click **New Project**

Select: Java Web → Web Application -> Click **Next**

Project name: PlayerREST

Click **Next**

Server: GlassFish Server

Click **Finish**

Project created ✅

6️⃣ Create REST Service

1. Right click **Project**

2. Click

New → RESTful Web Services from Database -> Click **Next**

7️⃣ Select Database

Choose:

Database Connection → playerDB

Select table: player

Click **Add ->** Click **Next**

8️⃣ Package Name

Enter: com.player.service

Click **Finish**

NetBeans will automatically generate:

Player.java
PlayerFacadeREST.java
ApplicationConfig.java
These files implement **CRUD automatically**.

9️⃣ Run the Project

Right click project -> Run

Browser will open.

Example URL: http://localhost:8080/PlayerREST

🔟 Test REST API

Open browser or Postman.

Insert Player (CREATE)

Method: POST

URL: http://localhost:8080/PlayerREST/webresources/player

Body (JSON):

{
"id":1,
"name":"Virat",
"country":"India"
}

Get Player (READ)

GET

URL: http://localhost:8080/PlayerREST/webresources/player/1

Output:

Virat

India

Update Player (UPDATE)

PUT

URL: http://localhost:8080/PlayerREST/webresources/player/1

Body:

{
"id":1,
"name":"Rohit",
"country":"India"
}

Delete Player (DELETE)

DELETE

URL: http://localhost:8080/PlayerREST/webresources/player/1

Player removed from database.

Practical 3 – JSP Client

Step 1 Create JSP File

Inside NetBeans project create a file: player.jsp

Step 2 JSP Code

Paste this code:

```
<%@page import="java.net.*"%>
<%@page import="java.io.*"%>
<html>
<head>
<title>Player REST Client</title>
</head>
<body>
<h2>Player Details</h2>
<%
try
{
   String url = "http://localhost:8080/PlayerREST/webresources/player/1";
   URL resturl = new URL(url);
   HttpURLConnection conn = (HttpURLConnection) resturl.openConnection();
   conn.setRequestMethod("GET");
   conn.setRequestProperty("Accept","application/json");
   BufferedReader br = new BufferedReader(
         new InputStreamReader(conn.getInputStream()));
   String output;
   while((output = br.readLine()) != null)
   {
      out.println(output);
   }
   conn.disconnect();
}
catch(Exception e)
{
   out.println(e);
}
%>
</body>
</html>
```

Step 3 Run JSP

Run project and open: http://localhost:8080/ProjectName/player.jsp

Example output:

{
"id":1,
"name":"Virat",
"country":"India"
}

What this JSP does

1. Connects to REST API
2. Calls **GET request**
3. Reads response
4. Prints player data

Libraries Used

No external libraries needed.
Only Java built-in packages:
java.net.*
java.io.*